

Malavika R

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PROFESSIONAL SUMMARY

Final-year Integrated M.Tech Software Engineering student specializing in Data Analytics and AI-driven systems. Experienced in developing intelligent applications using machine learning, deep learning, and natural language processing. Skilled in transforming raw datasets into meaningful insights and building data-driven solutions that integrate analytics with real-world AI systems.

PROJECTS

Speech Emotion Recognition for Tamil Conversations | NLP & AI Project

- Built a data-driven NLP system to analyze Tamil speech data and classify emotions such as happy, sad, and angry using machine learning and deep learning models.
- Processed and engineered audio features including MFCC, chroma, and spectral features using Librosa to extract meaningful patterns from raw speech signals.
- Developed CNN-based deep learning models and evaluated performance using accuracy, precision, recall, and confusion matrix analytics.
- Implemented a Flask-based AI application that predicts emotions from uploaded or recorded speech, demonstrating integration of data analytics with AI models.

Big Data Analysis and Demand Prediction For E-Commerce Products

- Built a Big Data analytics pipeline using Apache Spark and Trino to process and analyze large-scale Amazon product datasets stored in HDFS.
- Performed data preprocessing and feature engineering using PySpark, and executed high-performance SQL queries with Trino to extract insights such as category trends, ratings, and discount analysis.
- Developed a Linear Regression model to analyze relationships between price, discount, and ratings, enabling data-driven insights and predictive analysis for product performance.

Intelligent Video Surveillance for Violence Detection

- Developed an AI-powered system using CNN-LSTM architecture to detect violent activities from video streams.
- Applied VGG16 / MobileNetV2 with LSTM for spatiotemporal feature extraction and classification of violent and non-violent actions.
- Integrated Twilio API to send automated alerts to authorities upon detection of suspicious activities.

EDUCATION

Vellore Institute of Technology (VIT)

Integrated M.Tech in Software Engineering | **CGPA: 8.13**

2022 – 2027

Vellore, Tamil Nadu

V V N K M English Medium School

Higher Secondary Certificate | **Percentage: 73.8%**

2020 – 2022

Vellore, Tamil Nadu

V V N K M English Medium School

Secondary School Leaving Certificate | **Percentage: 79%**

2019 – 2020

Vellore, Tamil Nadu

SKILLS

Programming Languages: Python, Java, JavaScript, HTML, CSS

Core Concepts: SDLC, Data Structures and Algorithms, OOP, Agile

Data Analytics: Data Analysis, Data Preprocessing, Feature Engineering

Libraries & Tools: NumPy, Librosa, TensorFlow/Keras, Scikit-learn, Flask

AI Concepts: Machine Learning, Deep Learning, Natural Language Processing, Computer Vision

Soft Skills: Communication, Teamwork, Adaptability, Problem Solving

Languages: English, Tamil, Hindi